MCO-1D Complete Mode & Control Reference

Every wave library mode linked to its physical control · RadioMaster TX16S · Rev-H-A-B-C-D-E-F-G-H-I · 2026-06-07

CH3 — UNIFIED FLIGHT LEVER (right stick up/down) Full fwd = sprint · Full aft = max airbrake

Lever %	Zone	Freq Hz	Sweep °	Dihedral °	GL	Notes
0 – 15	SPRINT	2.5 → 3.0	135°	–8° anhedral	—	Max thrust / min drag. Sprint turn if CH1 deflected.
15 – 30	POWERED CRUISE	2.5 → 1.5	135°→90°	–8°→0°	—	Fast forward flight. Wings opening as speed reduces.
30 – 45	STD CRUISE	1.5 → 0.8	90°→60°	0°	—	Efficient cruise. Near best L/D speed.
47.5–52.5 ★	TRIM DETENT	0.2–0.4	60°	0°	GL-1	★ Hold altitude. 5% hysteresis — no hunting.
55 – 65	CLEAN GLIDE	0.2 → 0.0	60°	+0°→+8°	GL-0/1	Best glide. Motors off at 65%. Biological efficient glide.
65 – 75	APPROACH	0.3	60°	+8°→+18°	GL-2/3	Normal to steep approach. ~4°–5° glide angle.
75 – 85	STEEP DESCENT	0.3 → 0.5	60°→90°	+18°→+35°	GL-4/5	Full flap + speed brake equivalent. ~7°–11° glide.
85 – 95	NEAR-VERTICAL	0.5 → 0.8	90°→120°	+35°→+45°	GL-6	~25° descent angle. High dihedral airbrake.
95 – 100	MAX AIRBRAKE	0.8 → 1.0	120°→135°	+45°→+55°	GL-7	Maximum drag. ~30° descent. Emergency altitude loss.

AUTO MODIFICATIONS: mountain_mode adds +0.2 Hz bias and raises stall floor to Vs×1.25 (GPS terrain trigger). ground_roll/flare/touchdown automatically limit amplitude near ground.

CH1 — DIRECTION STICK (left stick L/R) Context via CH5 mode switch

CH5	Stick %	Mode triggered	Bank	Flapping	Notes
0 CRUISE	< 30%	MT-1 gentle turn	30°	0.4 Hz	Easy SAR manoeuvre
0 CRUISE	30–70%	MT-2 steep turn	35–45°	0.2 Hz	Standard turn
0 CRUISE	> 70%	MT-3 tight turn	55–60°	NONE	All lift into turn
0 CRUISE	100%	MT-4 max perf	65°	NONE	Altitude loss expected
1 MANOEUVRE	Any	MT-3/4 tight	45–65°	NONE	No flapping — biological
1 MANOEUVRE	≥ 50% + CH2 up	CHANDELLE macro	30°	2.2 Hz ramp	Climbing 180° turn — full macro
2 SAR	Left	ORBIT_LEFT	30°	0.5 Hz	CCW loiter over point
2 SAR	Right	ORBIT_RIGHT	30°	0.5 Hz	CW loiter over point
2 SAR	Centre	LOITER	0°	0.6 Hz	Hold GPS position

CH2 — PITCH TRIM (left stick U/D)

CH5	Input	Effect
0/1 FLIGHT	Up (+)	Adds +2°–+8° pitch offset — tightens turns, aids climb
0/1 FLIGHT	Down (–)	Adds –2° pitch — forward speed bias
0/1 FLIGHT	Centre	Pitch from sweep schedule (automatic)
2 SAR	Up	Steepens approach angle
2 SAR	Down	Shallows approach — longer final
1 MAN + CH1>50%	Up	Triggers CHANDELLE macro (climbing turn)

CH6 — WING CONFIG (3-pos switch SB)

Pos	GS	Active wings	Use
0	GS-0	W1+W2+W3+W4	Standard — all wings (default)
1	GS-1	W1+W2 only	Cruise glide — rear pairs stowed
1 + V>10 m/s	GS-2	W1 only	High-speed glide — 3 pairs stowed
2	GS-4	W1+W3	Slot glide (experimental)
2 + CH7	GS-3	W1+W4	Gap-wing glide (long press)
2 + CH7 long	GS-5	W3+W4	Rear-only (experimental)

CH4 — YAW (right stick L/R) ground only

State	Input	Effect
On ground	Left	Port amplitude up → turn left (ΔΦ*=0.30)
On ground	Right	Stbd amplitude up → turn right
In flight	Any	No effect (yaw via sweep asymmetry)

CH8 — SAFETY / RTH (2-pos switch SD)

Position	Action
0 (normal)	All controls active — normal flight
1 (RTH)	Return to home at cruise altitude
Flip 1→0 mid-macro	Abort running macro, return to cruise

CH7 — MACRO TRIGGER (momentary SF)

CH5	Press	Macro	Interruptible?
0 CRUISE	Short	CF-1 burst (J<0.8, V<10 m/s)	Yes
0 CRUISE	Long >1s	STOOP sequence	No (safety only)
1 MAN	Short	MT-4 max-perf turn	Yes
1 MAN	Long	CHANDELLE 180° climb	Yes
2 SAR	Short	SPIRAL DESCENT over GPS point	Yes
2 SAR	Long	APPROACH SITE final descent	Yes
Ground	Long	REVERSE taxi	Yes
Any	—	Abort: use CH8 flip	—

GROUND OPERATIONS (on ground / AGL < 0.15m)

Control	Ground function
CH3 lever fwd	Forward taxi speed (0=stop, 100%=fast)
CH4 yaw stick	Ground steering L/R (ΔΦ*=0.30 asymmetry)
CH6 = 2	INSPECTION MODE — wings to 90° abeam
CH7 long	REVERSE taxi (negative pitch rearward thrust)
CH8	Disarm (as normal)
CH5 context	Ignored on ground

AUTOMATIC MODES (no pilot input)

Mode	Trigger	What it does
ground_roll	AGL < 3m	Amplitude limiting begins
flare	AGL < 0.65m	Amplitude auto-reduces for landing
touchdown	AGL < 0.03m	Motors idle, wings stow W4→W1
stall_recovery	V < Vs×1.10	+0.3 Hz boost for 2s then resume
mountain_mode	GPS terrain ±500m	Freq +0.2 Hz, stall floor Vs×1.25
RECOVER	Stall imminent	Emergency: all wings 60°, freq 0.8 Hz
EMERGENCY_LAND	Battery < 19.8V	Immediate controlled descent

COMPLETE MODE INVENTORY — 37 MODES, ALL MAPPED

Mode	Control	Notes
`metachronal_90`	CH3 lever 30–50%	Standard cascade cruise
`metachronal_0 (sprint burst)`	CH3 lever 0–15%	High-speed synchronous
`CF-1 clap-and-fling`	CH7 short + CH5=0	Gated: J<0.8, V<10 m/s
`STOOP macro`	CH7 long + CH5=0	Locked; abort via CH8 flip
`chandelle macro`	CH7 long + CH5=1	180° climbing turn
`spiral_descent macro`	CH7 short + CH5=2	GPS-referenced descent
`stall_recovery`	Automatic	+0.3 Hz boost for 2s
`mountain_mode`	Automatic / GPS	Freq bias + raised stall floor
`GL-0 best glide`	CH3 ~60%	–5° anhedral, motors off
`GL-1 soaring`	CH3 ~65%	+8° dihedral, light flap
`GL-2 normal approach`	CH3 ~70%	+12° dihedral
`GL-3 steep approach`	CH3 ~75%	+18° dihedral
`GL-4 steep descent`	CH3 ~80%	+25° dihedral
`GL-5 very steep`	CH3 ~85%	+35° dihedral
`GL-6 near-vertical`	CH3 ~90%	+45° dihedral
`GL-7 max airbrake`	CH3 100%	+55° dihedral, 135° sweep
`GS-0 all wings`	CH6 = 0	Default, all 4 pairs
`GS-1 W1+W2`	CH6 = 1	Rear pairs stowed
`GS-2 W1 only`	CH6=1 + V>10 m/s	Auto-select at speed

Mode	Control	Notes
GS-3 W1+W4 gap	CH6=2 + CH7	Long press
GS-4 W1+W3 slot	CH6 = 2	Alternate pairs
GS-5 W3+W4 rear	CH6=2 + CH7 long	Experimental
MT-1 gentle turn 30°	CH1 < 30% + CH5=0	Light flapping
MT-2 steep turn 45°	CH1 30–70% + CH5=0	Minimal flapping
MT-3 tight turn 60°	CH1 > 70% + CH5=0/1	NO flapping — biological
MT-4 max-perf 65°	CH7 short + CH5=1	Maximum centripetal
MT-5 chandelle turn	CH1+CH2 up + CH5=1	Climbing tight turn
LOITER	CH5=2, CH1 centre	Hold GPS position
ORBIT_LEFT	CH5=2, CH1 left	CCW circle over point
ORBIT_RIGHT	CH5=2, CH1 right	CW circle over point
APPROACH_SITE	CH7 long + CH5=2	Final SAR descent
forward_taxi	CH3 on ground	Asymmetric stroke
reverse_taxi	CH7 long on ground	Negative pitch thrust
inspection	CH6=2 on ground	Wings to 90°
ground_roll/flare/touchdown	Automatic	AGL proximity
RTH	CH8 = 1	Return to home
RECOVER / EMERGENCY_LAND	Automatic	Stall / battery